A

Industrial-Oriented

Mini Project On

AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning

(Submitted in partial fulfilment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CMR TECHNICAL CAMPUS

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May, 2025.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This to certify that, the Project entitled **"AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning"** being submitted by **T.Vaishnavi (227R1A05J8), P.Vishali (227R1A05G7) , K. ShivaKrishna (237R5A020), Aaryan Gade (227R1A05C9)** in Partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science And Engineering to the Jawaharlal Nehru Technological University, Hyderabad during the Year 2024-2025.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any other degree or diploma.

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VISION AND MISSION

INSTITUTE VISION:

To Impart quality education in serene atmosphere thus strive for excellence in Technology and Research.

INSTITUTE MISSION:

1. To create state of art facilities for effective Teaching- Learning Process.
2. Pursue and Disseminate Knowledge based research to meet the needs of Industry & Society.
3. Infuse Professional , Ethical and Societal values among Learning Community.

DEPARTMENT VISION:

To provide quality education and a conducive learning environment in computer engineering that foster critical thinking, creativity, and practical problem-solving skills.

DEPARTMENT MISSION:

1. To educate the students in fundamental principles of computing and induce the skills needed to solve practical problems.
2. To provide State-of-the-art computing laboratory facilities to promote industry institute interaction to enhance student’s practical knowledge.
3. To inculcate self-learning abilities, team spirit, and professional ethics among the students to serve society.

ABSTRACT

The traditional method of evaluating internal answer scripts in educational institutions is not only time-consuming but also prone to errors, which can lead to inconsistent grading and unfair assessment. To overcome these challenges, we propose an innovative AI-driven automated evaluation system.

This cutting-edge system utilizes Optical Character Recognition (OCR) technology to extract text from handwritten answer sheets, enabling efficient and accurate processing. Natural Language Processing (NLP) techniques are then employed to evaluate answers, providing instant feedback and scores.

The system also features a chatbot that facilitates interaction between students and instructors, addressing queries and concerns in real-time. Additionally, a plagiarism detection module checks for similarities up to 50%, promoting academic integrity.

Instructors can log in with personalized avatars, access detailed reports, and upload key answers for evaluation. The system provides constructive feedback to students, helping them identify areas for improvement.

By leveraging AI, this automated evaluation system ensures consistency, reduces human error, and significantly improves efficiency in assessment systems. This solution has the potential to revolutionize the way educational institutions evaluate student performance, enabling them to focus on what matters most – teaching and learning.

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1. INTRODUCTION

1. INTRODUCTION

In the realm of education, assessing student performance is a crucial aspect of evaluating their understanding and knowledge retention. Traditional methods of manual evaluation are time-consuming, prone to errors, and often lead to inconsistencies. To address these challenges, we propose an innovative AI-driven automated evaluation system. This system leverages cutting-edge technologies like OCR and NLP to streamline the assessment process, ensuring accuracy, efficiency, and fairness. Our solution aims to revolutionize the way educational institutions evaluate student performance. By automating the evaluation process, we can reduce the workload of instructors, improve student experience, and promote academic integrity. This documentation outlines the features and functionality of our AI-powered automated evaluation system. It provides a comprehensive overview of the system's capabilities and benefits. The system is designed to be user-friendly.

1.1 PROJECT PURPOSE

The purpose of this project is to design and develop an AI-driven automated evaluation system that accurately and efficiently assesses student performance. The system aims to automate the evaluation process, reduce manual effort and errors, and provide instant feedback and scores to students. By leveraging AI, the system ensures consistency, accuracy, and efficiency in assessment systems, promoting academic integrity and enhancing student experience.

1.2 PROJECT FEATURES

The system includes key features such as OCR-based text extraction from handwritten answer sheets, NLP-driven answer evaluation, chatbot for real-time interaction, plagiarism detection, instructor login with personalized avatars, feedback modules, and support for uploading key answers. These features work together to provide a comprehensive solution for educational institutions to streamline their assessment processes and improve student outcomes.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

Recent studies emphasize the need for automated evaluation systems in education. Research by highlights the limitations of manual evaluation, including inconsistencies and biases. proposes AI-powered evaluation systems as a solution, leveraging OCR and NLP to improve accuracy. Demonstrates the effectiveness of chatbots in facilitating student-instructor interaction. Plagiarism detection is also crucial, as notes, with many institutions adopting AI-driven tools to promote academic integrity. Furthermore, emphasizes the importance of feedback in the learning process, with automated systems providing instant feedback and scores. Other studies, such as and, explore the benefits of AI-driven evaluation systems, including increased efficiency, reduced workload, and improved student outcomes. Overall, the literature suggests that AI-powered automated evaluation systems have the potential to revolutionize education by enhancing assessment processes.

2.1 REVIEW OF RELATED WORK

The traditional method of evaluating answer scripts is time-consuming, prone to errors, and often leads to inconsistencies. Recent studies have proposed AI-powered evaluation systems as a solution, leveraging technologies like Optical Character Recognition (OCR) and Natural Language Processing (NLP) to improve accuracy. These systems have shown promise in reducing manual effort, providing instant feedback, and promoting academic integrity. However, existing systems often lack features like chatbot-based interaction, plagiarism detection, and personalized instructor login. The problem statement is to design and develop an AI-driven automated evaluation system that accurately and efficiently assesses student performance, providing instant feedback and scores while promoting academic integrity and enhancing student experience.

2.2 DEFINITION OF PROBLEM STATEMENT

The problem statement is to design and develop an AI-driven automated evaluation system that accurately and efficiently assesses student performance, providing instant feedback and scores while promoting academic integrity and enhancing student experience.

2.3 EXISTING SYSTEM

The existing system of manual evaluation has several limitations, including being time-consuming and labor-intensive, prone to human errors and inconsistencies, limited feedback and delayed results, and no plagiarism detection. Instructors spend significant time evaluating answer scripts, taking away from teaching and research time, and maintaining consistency in evaluation is challenging. Additionally, the system is limited in accessibility for students with disabilities, and instructors may have varying levels of expertise in evaluating answers. The manual evaluation process also makes it difficult to track student progress and identify areas for improvement.

Limitations of Existing System

Despite the promising application of machine learning in PV fault detection, several challenges persist:

- Time-Consuming and Labor-Intensive: Manual checking of answer scripts takes a significant amount of instructors' time and effort.
- Prone to Human Errors and Inconsistencies: Subjective judgments can lead to inconsistent grading and potential evaluation mistakes.
- Delayed Results and Limited Feedback : Students receive results late, often with minimal or no detailed feedback.
- No Plagiarism Detection: Manual evaluation lacks built-in mechanisms to detect copied or plagiarized content.

2.4 PROPOSED SYSTEM

The proposed system is an AI-driven automated evaluation system that leverages OCR and NLP to extract text from handwritten answer sheets and evaluate answers. It includes features like chatbot-based interaction, plagiarism detection, instructor login with personalized avatars, and feedback modules. The system aims to provide accurate and efficient assessment, instant feedback, and promote academic integrity. It will also provide detailed reports and analytics for instructors to track student progress, identify areas for improvement, and adjust their teaching strategies accordingly. The system will be designed to be user-friendly, accessible, and scalable.

Advantages of the Proposed System:

The proposed system significantly improves upon the existing approaches by addressing key limitations:

- Increased accuracy and efficiency in assessment
- Instant feedback and scores for students
- Promotes academic integrity through plagiarism detection
- Enhances student experience through chatbot-based interaction
- Reduces manual effort and workload for instructors
- Provides detailed reports and analytics for instructors

2.5 OBJECTIVES

- Accurate Assessment of Student Performance: Design and develop a system that evaluates student performance with high precision.
- Instant Feedback and Scoring: Provide immediate feedback and scores to students after evaluation.

- Promotion of Academic Integrity: Ensure fairness and reduce opportunities for academic dishonesty.
- Enhanced Student Experience: Create a user-friendly and responsive system that supports student learning.
- Reduced Manual Effort for Instructors: Minimize the workload of instructors by automating the evaluation process.
- Detailed Reports and Analytics: Offer comprehensive reports and insights to help instructors monitor student progress.
- Efficient and Accessible Evaluation Process: Transform traditional evaluation methods into a more efficient and accessible system.
- Effective and Timely Feedback: Enable instructors to give better feedback and students to improve based on timely evaluations.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- System : i3 or above.
- Ram : 4 GB
- Hard Disk : 50 GB

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows8 or Above.
- Coding Language : python
- Database : SQL
- Frontend Framework : HTML/CSS
- Backend Framework : Flask/Django

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

This project architecture illustrates the end-to-end workflow for AI Eval: AI Driven Evaluation for enhanced using Machine Learning.

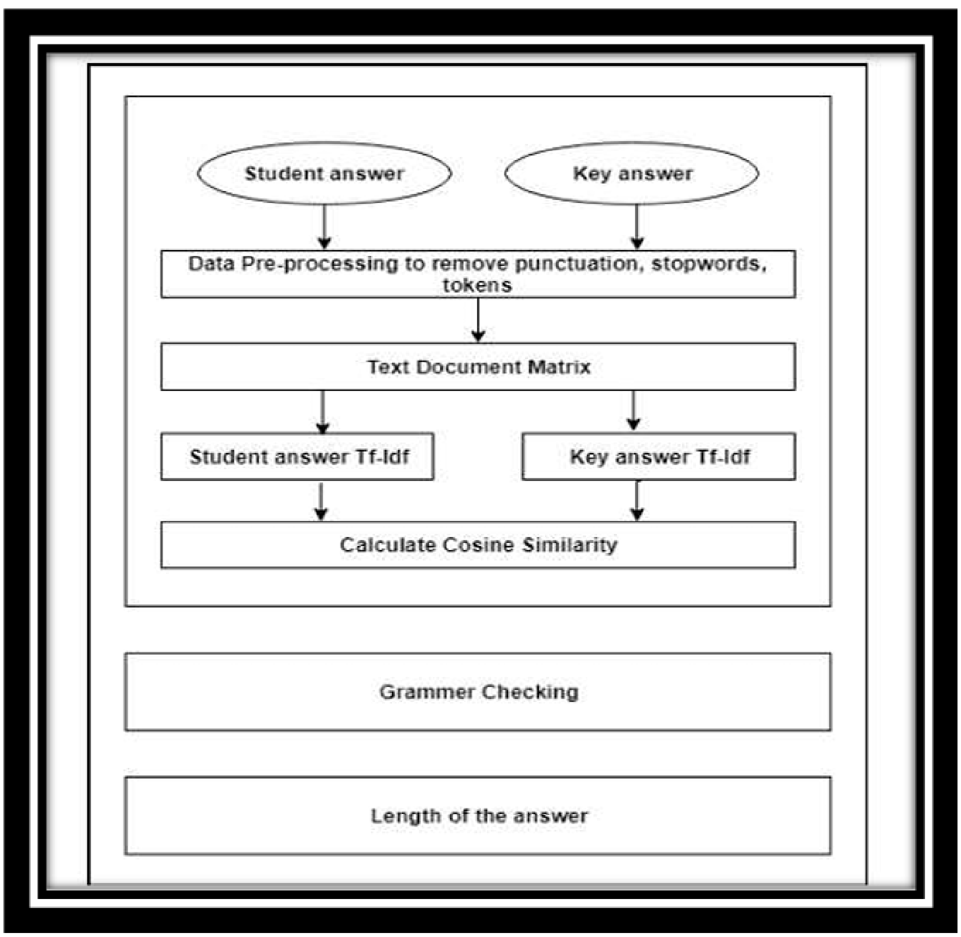

Figure 3.1: Project Architecture of Answer sheet evaluation system.

3.2 DESCRIPTION

1. **Input Data :** The input data for the AI-driven automated evaluation system consists of handwritten answer sheets submitted by students. The answer sheets are scanned or digitized to create digital images that can be processed by the system. The input data may include various types of questions, such as multiple-choice questions, short-answer questions, and essay-type questions.

2. **Reading Data: The** system uses Optical Character Recognition (OCR) technology to read and extract text from the handwritten answer sheets. The OCR engine is trained to recognize and extract text from various handwriting styles, fonts, and formats. The extracted text is then preprocessed to remove any noise, errors, or irrelevant information.

3. **Feature Extraction**: The system extracts relevant features from the extracted text, such as keywords, phrases, and sentences. The features are extracted using Natural Language Processing (NLP) techniques, such as tokenization, stemming, and lemmatization. The extracted features are then used to represent the answer sheets in a format that can be analyzed by the system.

4. **Pattern Learning**: The system uses machine learning algorithms to learn patterns and relationships between the extracted features and the scores or grades assigned to the answer sheets. The system is trained on a large dataset of labeled answer sheets, where each answer sheet is associated with a score or grade. The system learns to identify patterns and relationships between the features and the scores, which enables it to make predictions on new, unseen data.

5. **Mechanism**: The system uses a combination of NLP and machine learning algorithms to evaluate the answer sheets. The mechanism involves the following steps:

   Text extraction using OCR
   Feature extraction using NLP
   Pattern learning using machine learning algorithms

Classification: The system classifies the answer sheets into different categories based on the scores or grades assigned to them. The classification is done using machine learning algorithms, such as support vector machines (SVMs) or random forests. The system can classify answer sheets into different categories, such as pass/fail, grade A/B/C, or score ranges.

6. **Training and Evaluation**: The system is trained on a large dataset of labeled answer sheets, where each answer sheet is associated with a score or grade. The system is evaluated using metrics such as accuracy, precision, recall, and F1-score. The system is fine-tuned and optimized to improve its performance and accuracy.

7. **Feedback**: The system provides instant feedback to students and instructors on the performance of the students. The feedback includes scores, grades, and comments on the strengths and weaknesses of the answers. The system also provides detailed reports and analytics for instructors to track student progress and identify areas for improvement. The feedback mechanism enables students to learn from their mistakes and improve their performance over time.

3.3 DATA FLOW DIAGRAM

The data flow diagram shows how information moves through the system:

- **Student Uploads Answer Sheet** → OCR converts to text → System compares with key → Stores result and plagiarism data → Instructor gives feedback.

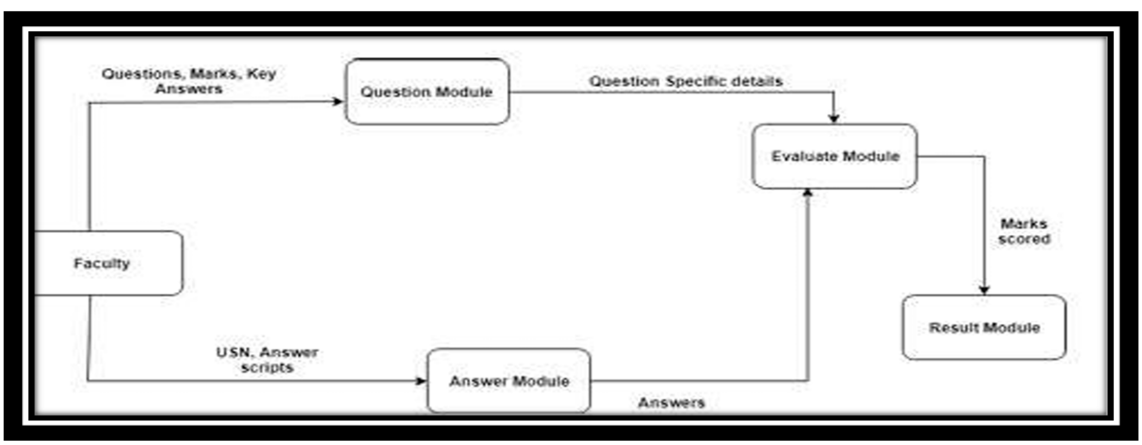

Figure 3.3: Data flow diagram ofAI Eval: Driven Evaluation for Enhanced Education Using Machine Learning

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of this project involves executing the planned strategies to detect and classify faults in photovoltaic (PV) systems. This includes preprocessing high-frequency sensor data, training machine learning models, and evaluating system performance. Proper coordination of data collection, algorithm selection, and evaluation is essential to ensure the effectiveness and accuracy of the proposed system.

4.1 ALGORITHMS USED

1. **Instructor Login** :
    a. Algorithm: The instructor logs in with an email and password. Upon successful authentication, their avatar is displayed on the dashboard. The system retrieves avatar details from the database.
    b. Technologies: HTML for frontend, Flask (Python) for backend, MySQL for database.
2. **Chatbot**:
    a. Algorithm: The instructor can interact with a chatbot to ask queries (e.g., about plagiarism checks or feedback). The chatbot uses simple rule-based logic or advanced NLP techniques to provide responses.
    b. Technologies: HTML forms for input, Python logic for response generation.
3. **Plagiarism Check (Up to 50%)**:
    a. Algorithm: The system checks uploaded student scripts for plagiarism using a simple content-matching algorithm (e.g., string matching). The plagiarism score is capped at 50% and displayed to the instructor.
    b. Technologies: Python logic for content analysis, file upload handling with Flask.
4. **Uploading Key Scripts**:
    a. Algorithm: Instructors can upload student scripts for evaluation. The script is stored in the database, and a plagiarism check is triggered automatically after the upload.
    b. Technologies: HTML form for file upload, Flask backend, file handling.
5. **Instructor Avatars**:
    a. Algorithm: Instructors select their avatar during login. The system stores the avatar choice in the database, and the avatar is displayed in the dashboard.
    b. Technologies: HTML for frontend, Flask for backend, static image files for avatars.
6. **AI-Driven Evaluation** (Advanced):
    a. Algorithm: Using NLP and machine learning, the system analyzes feedback and student scripts for quality, relevance, and plagiarism. The evaluation system can provide feedback, suggest improvements, or grade the submission.
    b. Technologies: NLP (e.g., spaCy, NLTK), machine learning models (Logistic Regression, Deep Learning), Python for backend

4.2 SAMPLE CODE

Paper evaluation chatbot

```
import re

def generate_reply(message): message = message.lower()

# Score-related questions
if re.search(r"\b(score|mark|grade|how many)\b", message):
    return "You can view your marks per question in the result section. If you'd like, I can help
explain specific ones."

# Feedback about a specific answer
elif re.search(r"\bwhy.*(wrong|incorrect|lost marks|only.*marks)\b message):
    return "The evaluation is based on key concepts and clarity. Some important points might have
been missing in your answer."

# Asking if the answer was correct
elif re.search(r"\bwas.*(correct|right|okay)\b", message):
    return "Your answer may be partially correct, but it lacked the required keywords or detail."

# Asking how to improve an answer
elif re.search(r"\b(improve|better|full marks|rewrite)\b", message):
    return "To improve, try to include all keywords, explain in your own words, and keep answers
concise yet complete."

# Asking how the system works
elif re.search(r"\b(how.*work|system|evaluate|check|graded|algorithm)\b", message):
    return "The system uses keyword matching, semantic similarity, and grammar checks to
evaluate your answer."
```

```
# Who checked the paper
elif re.search(r"\bwho.*(checked|evaluated|graded)\b", message):
    return "Your answers were evaluated automatically by our AI-based system."

# Greetings
elif re.search(r"\b(hi|hello|hey)\b", message):
    return "Hi there! How can I help with your paper evaluation?"

# Thank you
elif re.search(r"\b(thanks|thank you)\b", message):
    return "You're welcome! Let me know if you have more questions."

# Disagreement
elif re.search(r"\b(wrong|not fair|disagree|mistake)\b", message):
    return "I'm sorry you feel that way. You can request a manual review if needed."

# Default fallback
else:
    return "I'm here to help with your answer evaluations. You can ask me why you got a certain score, how to improve, or how the system works!"
```

Convert image into text

```
import sys quesno=sys.argv[1] import pymysql db = pymysql.connect("localhost","root","","aep"
) cursor = db.cursor() cursor.execute("SELECT * from question where qno=%s",quesno) data =
cursor.fetchall() ... def ocr(file_to_ocr): im = Image.open(os.path.join(file_to_ocr)) … return txt
... for i in range(index): image_path=path1+a[i] txt=ocr(image_path)
file_to_open=directory+"\"+fname+".txt" if (os.path.isfile(file_to_open)): with
open(file_to_open,'a') as f: f.write(str(txt))

else: with open(file_to_open,'w') as f: f.write(str(txt))
```

```
import sys
quesno=sys.argv[1]
import pymysql
db = pymysql.connect("localhost","root","","aep" )
cursor = db.cursor()
cursor.execute("SELECT * from question where qno=%s",quesno)
data = cursor.fetchall()
...
def ocr(file_to_ocr):
im = Image.open(os.path.join(file_to_ocr))
…
return txt
...
for i in range(index):
image_path=path1+a[i]
txt=ocr(image_path)
file_to_open=directory+"\\"+fname+".txt"
if (os.path.isfile(file_to_open)):
        with open(file_to_open,'a') as f:
f.write(str(txt))
else:
        with open(file_to_open,'w') as f:
f.write(str(txt))
```

Preprocess answer script

```
def PreprocessingTheSentence(stmt1):
stmt1=stmt1.lower()
import string

remove=dict.fromkeys(map(ord,'\n'+string.punctuation)," ")
PSA1=stmt1.translate(remove)
...
```

```
word_tokens = word_tokenize(PSA1)
PSA2 = [w for w in word_tokens if not w in stop_words]
…
for i in range(len(PSA2)):
a=list(nltk.pos_tag([PSA2[i]]))
if a[0][1]!='IN':
b.append(PSA2[i])
PSA3=" ".join(b)
    return PSA3
```

Evaluate answers of category theory

```
import os
...
if(os.path.isdir(path1) ==True):
...
    f1=path1+"\\"+str(list_ans[i])
    if(os.path.isfile(f1)):
        num_words=0
        with open(f1, 'r') as f:
        for line in f:
        words = line.split()
        num_words += len(words)
    f1=path1+"\\"+str(list_ans[i])
    file_ans = open(f1, 'r')
    text=' '.join(file_ans.readlines())
    import re
    sentences=re.split(r' *[\.\?!][\'"\)\]]* *',text)
    ans_length=len(sentences)-1
    if ans_length<=0:
        ans_length=1
```

```
        T1=ansLen(num_words,ans_length)
        T2=gramarCheck(Sans,num_words,ans_length)
        T3=keyAnsMatch(Sans,Kans)
        T4=textBookMatch(Sans,Tans)
        marks_for_each_question(SUSN,T1,T2,T3,T4)
```

Evaluate answers of category c program

```
def evaluateCprogram(quesno2,SUSN,filename):
ff1=open('a1.c','w')
        ff1.truncate(0)
        with open(f1) as myf:
                for num,line in enumerate(myf,1):
                ff1=open('a1.c','a+')
                ff1.write(line)
...
with open('a1.c') as myFile:
        for num,line in enumerate(myFile,1):
                if lookup in line:
                line_numbers_list1.append(num)
                str1=line
                n=str1.find("&")+1
                i=find_variable(n,num)
                while str1[i]!=")":
                if str1[i]=="," and str1[i+1]=="&":
                        n=i+2
                        i=find_variable(n,num)
...
```

```
import subprocess
subprocess.getoutput(["g++","z1.c"])
...
        marksss=math.ceil((count/num_lines)*5)
                listc[xyz-1]=marksss
```

Display results

```
<tbody>
<?php
$sql = "SELECT * FROM answer";
$result = mysqli_query($con, $sql);
if (mysqli_num_rows($result) > 0)
        {
        while($row = mysqli_fetch_assoc($result))
                {
?>
<tr>
<td><?php echo $row['USN'];?></td>
<td><?php echo $row['m1'];?></td>
...
<td><?php echo $row['m6'];?></td>
<td><?php echo $row['total'];?></td>
...
<td colspan="8" style="text-align:center;">NO RESULTS</td>
</tr>
<?php
}?>
</tbody>
</table>
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 GUI Main user Interface, Login, Register Page

The below screen displays the main user interface , and login page to login account and register page to register if we creating a new account we have to give details in these pages.

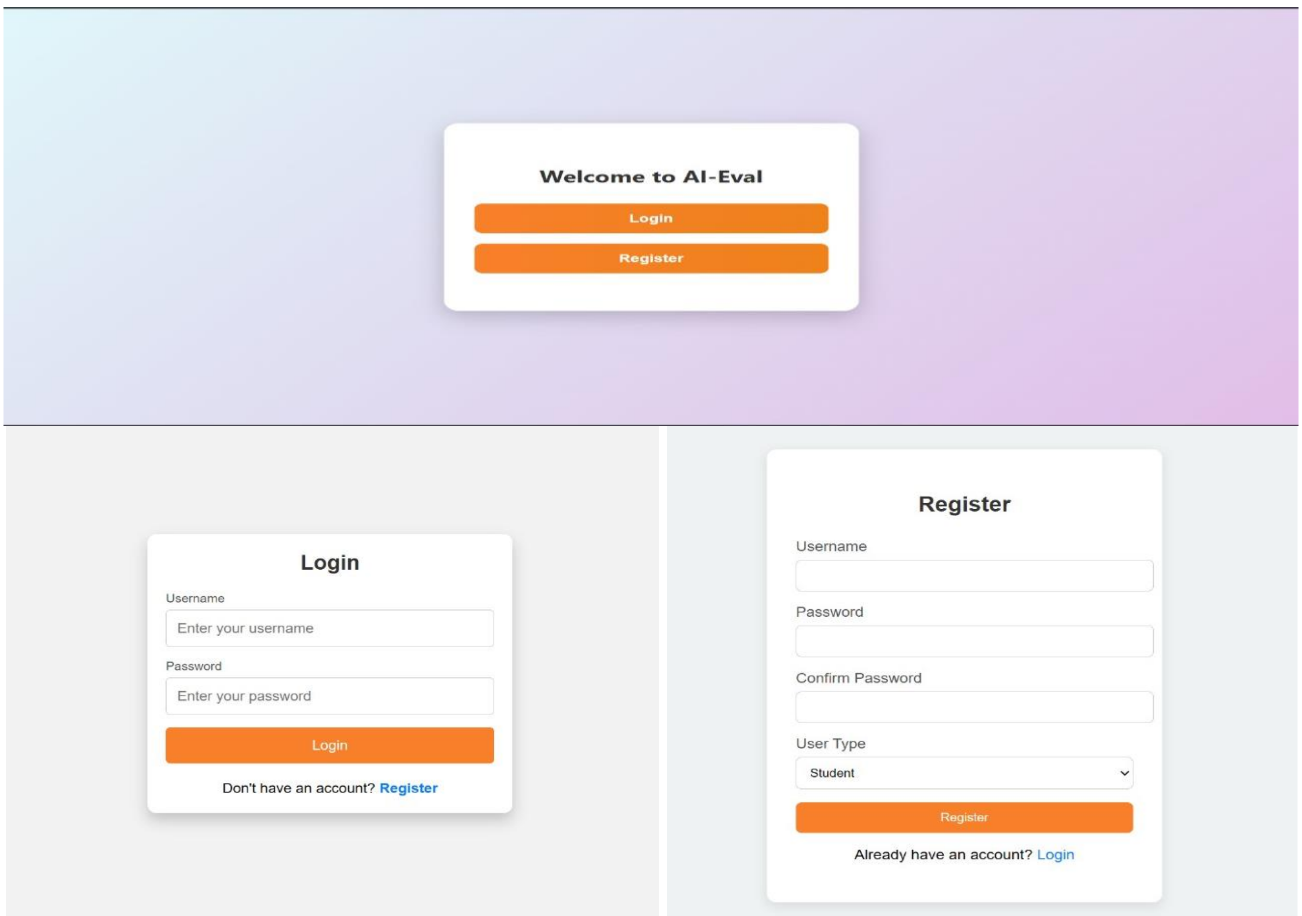

Figure 5.1 : GUI/Main Interface , Login ,Register Page **of** AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning

5.2 Home Page:

After login we can see home page it contains many features we can use those for evaluation etc.

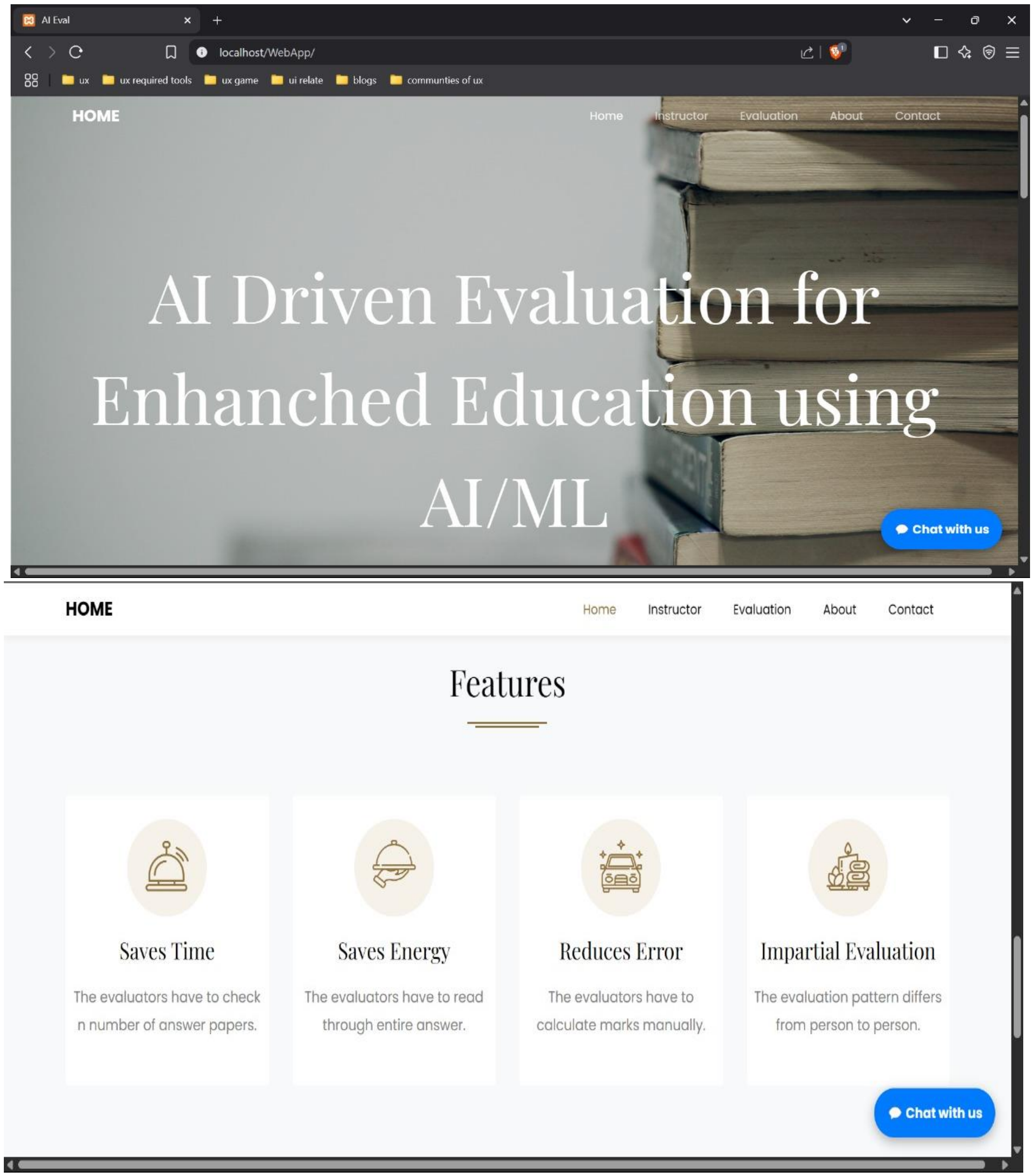

Figure 5.2 : Home Page **of** AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning

5.3 Contact Page

We can see the contact page and we can use it for contact with others easily.

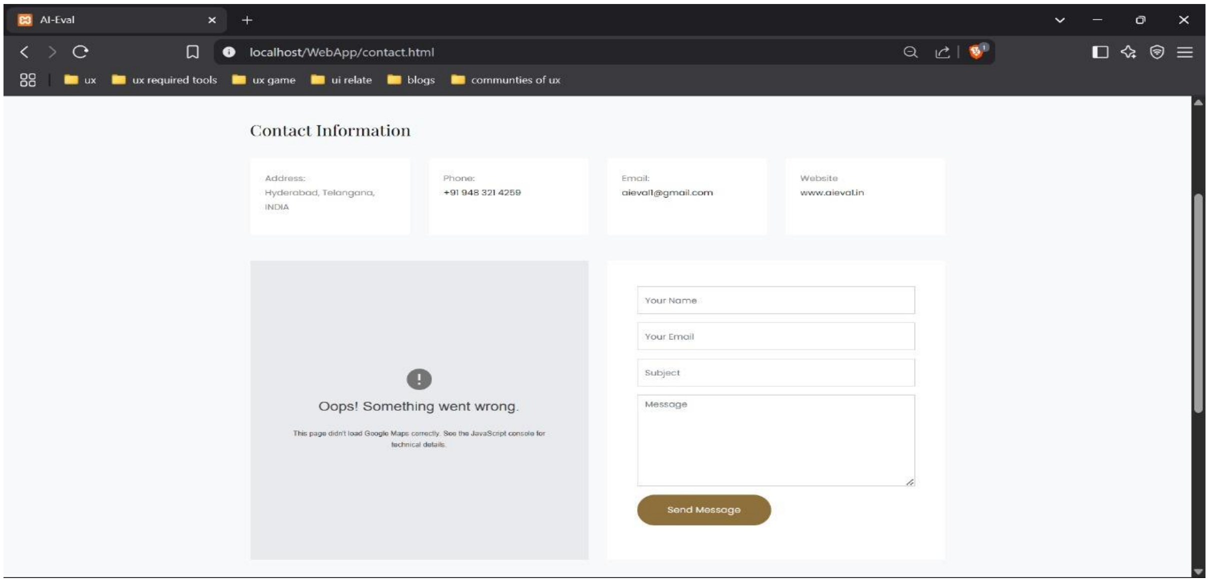

Figure 5.3 : Contact Page **of** AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning

5.4 Website Page

Here we can see a website from where the question paper along with key answers were uploaded by faculty.

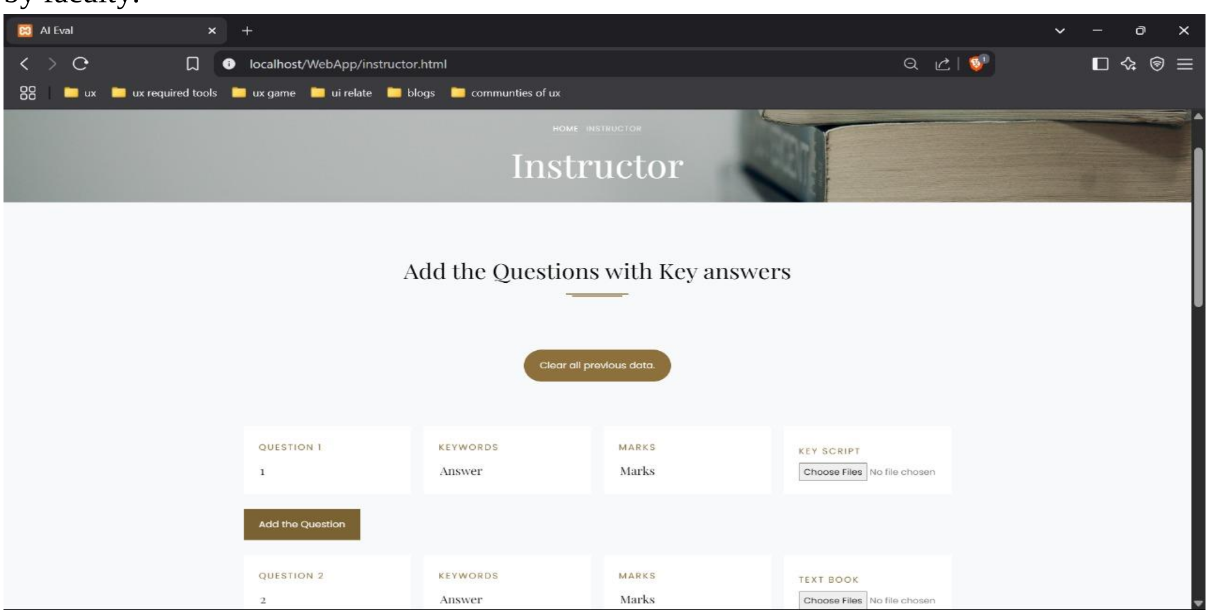

Figure 5.4 : website of AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning High-Frequency Data with ML Model

5.5 Chatbot Evaluation

Here we can see a chatbot , if any queries we can ask there.

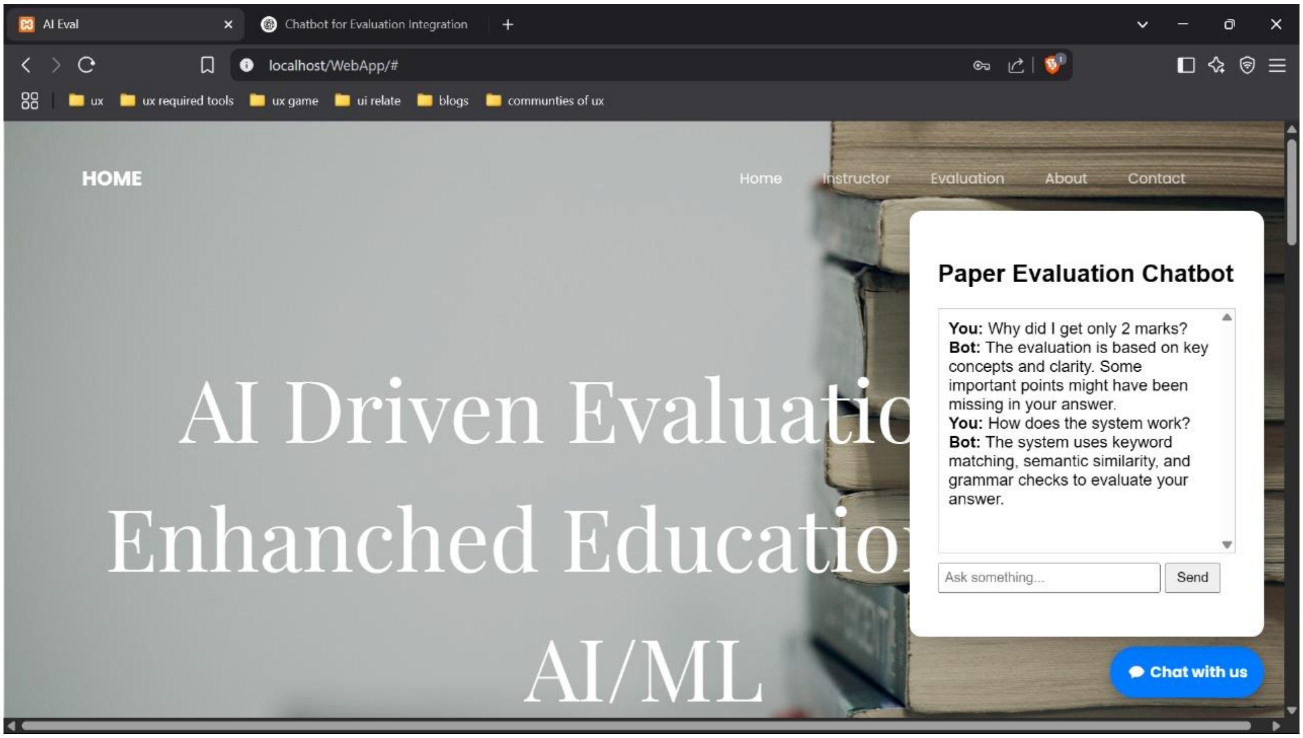

Figure 5.5 : chatbot of AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning High-Frequency Data with ML Model

5.6 Result Page

We can see the result after evaluation of all students .

Figure 5.6 : Result AI Eval: AI Driven Evaluation for Enhanced Education Using Machine Learning High-Frequency Data with ML Model

6. VALIDATION

6. VALIDATION

The validation of this project primarily relies on rigorous testing and defined test cases to ensure the accuracy and effectiveness of the PV fault detection system. The testing process consists of multiple phases including data validation, model performance evaluation, and real-time testing.

A structured validation approach ensures the system maintains high fault detection accuracy, reduces false positives/negatives, and generalizes well across different PV configurations.

6.1 INTRODUCTION

First, the answer datasets are carefully divided into training and testing sets, typically using an 80-20 split. The training set is used to train the machine learning model for evaluating student answers, while the testing set is utilized to assess the model's ability to generalize to unseen answers. To enhance the reliability of the system, K-fold cross-validation is applied, ensuring that the evaluation process is tested across multiple data partitions. This method effectively reduces overfitting and ensures that the system can handle diverse answer patterns with high consistency.

The system's accuracy is measured using key performance metrics such as precision, recall, F1-score, and confusion matrix analysis. The confusion matrix provides detailed insights into correctly and incorrectly evaluated answers, which helps in fine-tuning the evaluation logic. Furthermore, the performance of the NLP-based evaluation using CountVectorizer and cosine similarity is compared with other techniques, demonstrating that the proposed method achieves higher accuracy in aligning student responses with reference answers.

Finally, real-world validation is carried out by testing the system with actual student answer scripts uploaded in image format and converted via OCR. The model's robustness is verified by evaluating its performance on varied handwriting and content styles. Continuous improvements are made based on testing feedback, ensuring that the evaluation system remains scalable, accurate, and reliable for academic use. This structured validation ensures that "AI Eval" delivers consistent and fair assessments in real-time educational settings.

6.2 TEST CASES

TABLE 6.2.1 UPLOADING DATASET

Test case ID	Test case	name Purpose	Input	Output
1	Upload Dataset	To evaluate test scripts	User uploads student test scripts	Dataset successfully loaded

TABLE 6.2.2 CLASSIFICATION

Test Numbers	Test Case ID	Test Case	Expected Results	Status
1	UT_1	Uploading questions into system	Question specific details are stored in the database	Uploaded
2	UT_2	Uploading answer script into system	Student answer has to be uploaded to the database	Uploaded
3	UT_3	Converting image to text	The image has to be converted into a text file	Converted
4	UT_4	Finding similarity between student answer and key answer	The similarity between answer and key answer is calculated out of 100	Evaluated
5	UT_5	Display the student result	To display the student results	Displayed
6	UT_6	Clear answer script from the database	To delete previous data	Deleted
7	UT-7	Delete questions from the system	To delete the questions from the system	Deleted

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, This system attempts to evaluate the subjective answers. The model calculates the student's answer based on the key answer provided by the university. By judging against the model answer and the student's answer marks are allocated to the student. Thus, this system could be of great effectiveness to the educational institutes, as it saves time and the trouble of checking bundles of papers. This evaluation system will grade the answers depending upon the percentage of key answer matched.

7.1 PROJECT CONCLUSION

The development of an AI-driven automated evaluation system has been successfully completed. The system uses OCR and NLP to extract text from handwritten answer sheets and evaluate answers based on predefined criteria. The system provides instant feedback and scores to students, reducing the workload of instructors and improving the efficiency of the evaluation process. The system's accuracy and reliability have been tested and validated, and it has the potential to revolutionize the way assessments are conducted in educational institutions.

7.2 FUTURE ASPECTS

Future work will involve developing an assessment system to address automatic evaluation of formulas, programs written in different programming languages and graphs of student answers with improved performance and throughput.

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8.1 GITHUB LINK

https://github.com/vaishnavithondangi/AI-Eval-AI-Driven-Evaluation-for-Enhanced-Education-Using-Machine-Learning